COLLEGE OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CMAI 480 Computer Vision

FALL 2026

Instructor Information

Name: Dr. Khalil Yousef
Academic title: Visiting Professor
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Office Hours: Female Sections: Sunday, 12 pm – 12:50 pm, H07 – A399
Male Sections: Tuesday, 1 pm – 1:50 pm, H07 – B114

TA Information

TBA

Class/Laboratory Schedule

16495 - L01 - 02:00 pm - 02:50 pm H07-C111

16503 – L51 - 03:00 pm - 03:50 pm H07-A243

16504 – L52 - 04:00 pm - 04:50 pm H07-C130

Coordinator Information

Dr. Uvais Qidwai/ uqidwai@qu.edu.qa / H07-C337

Course Information

Catalog Description:
Principles and techniques of computer vision, including image formation, image processing, feature detection and matching, image alignment and stitching, and camera calibration. The course introduces fundamental deep learning methods such as convolutional neural networks (CNNs) for image classification, object detection, and semantic segmentation. Advanced topics such as Vision Transformers and Generative Adversarial Networks (GANs) may be surveyed as time allows. Students gain hands-on experience designing and implementing computer vision systems for real-world applications.

Contact Hours:
3 Hours Theory.

Prerequisites:

Data structures, Python/C++/MATLAB programming, Basics of mathematics.

Textbook(s):

"Computer Vision: Algorithms and Applications" by Richard Szeliski which can also be found online: http://szeliski.org/Book

References:

- Richard Hartley and Andrew Zisserman, "*Multiple View Geometry in Computer Vision*", Cambridge University Press, 2004, Second Edition, (https://www.robots.ox.ac.uk/~vgg/hzbook/).
- David A. Forsyth and Jean Ponce, "*Computer Vision: A Modern Approach*"; Prentice Hall, 2003. Electronic version available at: http://www.cs.berkeley.edu/~daf/book3chaps.html.
- Rafael C. Gonzalez, Richard E. Woods, *Digital Image Processing; 3rd Edition;* Prentice
- Hall; August 31, 2007.L. Shapiro and G. Stockman, *Computer Vision*, Addison Wesley, 2001.
- Mubarak Shah, "*Fundamentals of Computer Vision*"; http://www.cs.ucf.edu/courses/cap6411/book.pdf
- This web site lists a large set on CV books available online http://homepages.inf.ed.ac.uk/rbf/CVonline/books.htm
- Articles as determined by the instructor.
- Image Processing Toolbox, online reference and tutorials, (https://www.mathworks.com/products/image-processing.html).
- "Computer Vision Toolbox reference and tutorials (https://www.mathworks.com/products/computer-vision.html)
- Deep Learning Toolbox, online reference and tutorials, (https://www.mathworks.com/products/deep-learning.html)
- OpenCV - Open Computer Vision Library (https://opencv.org/).
- Several web resources

Course Objectives:

This course uses Project Based Learning (PBL) to introduce the fundamentals of Computer Vision, covering basic image acquisition systems and the underlying mathematics for representation and modeling of visual data. Students study image processing tailored for computer vision applications, using conventional as well as AI-based algorithms, gaining hands-on experience by completing a course project designed to solve real-life problems in the surrounding community.

- An overview of the computer vision field and its core principles.
- An understanding of basic methods, procedures, algorithms, and applications of computer vision using conventional and AI-based approaches.
- Hands-on experience in coding computer vision algorithms.
- Practical application of computer vision techniques to solve real-world problems in the surrounding community.

Course Learning Outcomes (CLOs):

1. Understand and master basic knowledge, terminology, mathematics, models and methods in the field of computer vision.
2. Identify, formulate and solve problems in computer vision through appropriate assessment and selection from the existing methods.
3. Design and develop practical and innovative image processing and computer vision applications or systems.

Relationship of Course Outcomes to Student Outcomes:

Course Learning Outcomes (CLO)	Related CE Student Outcomes (SO)						
	1	2	3	4	5	6	7
1	√						
2		√					
3						√	

Course Learning Outcomes (CLO)	Related CS Student Outcomes (SO)					
	1	2	3	4	5	6
1	√					
2		√				
3						√

CE Student Outcomes (CE-SO):

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. An ability to communicate effectively with a range of audiences.
4. An ability to recognize ethical and professional responsibilities in engineering

situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

CS Student Outcomes (CS-SO):

1. Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.
2. Design, implement and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.
3. Communicate effectively in a variety of professional contexts.
4. Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.
5. Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.
6. Apply computer science theory and software development fundamentals to produce computing-based solutions.

Week(s)	**Topics**	**Chapter/Section(s)**	**Weeks**
23/08/26 W 1	Introduction to Computer Vision (CV)	1.1	1
30/08/26 – 13/09/26 W 2&3&4	Image Formation Assignment (1) – Due End of W3	2.1,2.2,2.3	3
20/09/26 W5	Image Processing Assignment (2) – Due End of W5	3	1
27/09/26 – 04/10/26 W 6 &7	Feature Extraction and Matching Assignment (3) – Due End of W7 Midterm Exam – W 7: includes all course material covered up to (**but not including**) Feature Extraction and Matching	7	2
11/10/26 – 18/10/26 W8&9	Image Alignment and Stitching Assignment (4) – Due End of W9	8	2
25/10/26 W 10	Mid-Fall Break	-	1

1/11/26 W 11	Introduction about camera calibration	11.1	1
8/11/26 – 15/11/26 W 12 & 13	Fundamentals of machine-learning and Deep Learning on Course Project: Classification and Semantic Segmentation	4,5	2
22/11/26 W 14	Projects Presentations	-	1
29/11/226 W 15	Dead Week - Review (Last day of classes - 3 December 2026)	-	1
Total Week			15
Start of final exams (Fall 2026) 6 December 2026 End of final exams (Fall 2026) 17 December 2026			

Method of Instruction

In class Lectures (50 minutes), 3 times /week

Assessment Methods and Grading Policy

- Assignments: 15%
- Project: 35%
- Midterm Exam: 20% (TBA)
- Final Exam: 30%

ABET Contribution of Course to Professional Component

Subject Area (% Contribution)

- Math & Basic Science: 30%
- Engineering : 30%
- Engineering Design : 30%
- Broad Education : 10%

Computer/Software Usage

- Open Source software: (a) Python, (b) CV2 Library
- MATLAB (University License) as needed

Course Project

This is a Project-Based Learning (PBL) course centered around a hands-on group project worth 35% of the total grade. Students will work in teams of five to select and complete a project from a curated list, focusing on solving a real-world problem. The project involves designing and implementing a small-scale computer vision application within the MATLAB environment. Implementations must utilize either real-world data from public repositories or live image acquisition via suitable computer hardware. Progress will be evaluated through scheduled

milestones throughout the semester, culminating in a formal final report and a live demonstration of the working system.
Additionally, four biweekly assignments will be given to reinforce key course concepts. MATLAB will serve as the primary programming language for both assignments and the project. All guidelines, deadlines, and submission portals will be made available on Blackboard.

Course Ground Rules

Course website: http://mybb.qu.edu.qa (the Blackboard system) must be regularly consulted for assignments/updates and lectures.

Class Attendance: Students are required to attend classes on a regular basis. Attendance will be verified in every class meeting during the first 5 minutes only. The university regulations for class attendance are to be observed. It is the student's responsibility to find out about all the activities that take place in class on his/her absent days, including assignments, announcements and lecture materials. Students must present a valid excuse for absences for which they wish to receive an official absence excuse. Read the Undergraduate Student Handbook about the attendance policy of QU in page 29. It states: "A student should not miss more than 25% of the classes during a semester. Those exceeding this limit will receive a failing grade regardless of their performance."

http://www.qu.edu.qa/students/documents/handbook/undergraduate_student_handbook_english.pdf

Academic Honesty: Conventional or AI-Generated Plagiarism will not be tolerated. The academic dishonesty will be handled according to Qatar University policies. Copying and submitting someone else work will result in zero grade for all students who are involved in the action (the ones who copied and the one who allows his work to be copied).

University Code of Conduct

QU expects its students to adopt and abide by the highest standards of conduct in their interaction with professors, peers, staff members and the wider university community. Moreover, QU expects its students to act maturely and responsibly in their relationships with others. Every student is expected to assume the obligations and responsibilities required from them for being members of the QU community.

As such, a student is expected not to engage in behaviors that compromise their integrity, as well as the integrity of QU. Further information regarding the University Code of Conduct may be found on the web at http://www.qu.edu.qa/students/code-of-conduct

Use of Artificial Intelligence (AI) Tools

In accordance with Article 6 of the Student Code of Conduct at Qatar University, academic violations include a range of actions, one of which pertains to submitting work that is not the individual's own production. This includes using creative artificial intelligence tools such as ChatGPT to produce content, images, videos, or programming code and presenting it as original work.
Therefore, students are cautioned that using artificial intelligence tools such as ChatGPT or any similar tools to produce academic content and present it as their own work is considered plagiarism, exposing the student to disciplinary penalties as stipulated in Qatar University's Student Code of Conduct. In light of this, we urge all students to adhere to ethical standards in all assignments and

academic work, and to seek guidance from the course instructor when unsure about the proper and ethical use of artificial intelligence sources in completing assignments, duties, and academic tasks.

Support for Students with Special Needs

It is Qatar University policy to provide educational opportunities that ensure fair, appropriate and reasonable accommodation to students who have disabilities that may affect their ability to participate in course activities or meet course requirements. Students with disabilities are encouraged to contact their Instructor to ensure that their individual needs are met. The University through its Inclusion and Special Needs Support Center will exert all efforts to accommodate for individuals' needs.

Contact Information for Inclusion and Special Needs Support Center:
Tel-Female: (00974) 4403 7972
Tel-Male: (00974) 4403 7946
Location: Student Activities Building
Email: specialneeds@qu.edu.qa

Academic Support and Learning Resources

The University Student Learning Support Center (SLSC) provides academic support services to male and female students at QU. The SLSC is a supportive environment where students can seek assistance with academic coursework, writing assignments, transitioning to college academic life, and other academic issues. SLSC programs include: Peer Tutoring, the Writing Lab, Writing Workshops, and Academic Success Workshops. Students may also seek confidential academic counseling from the professional staff at the Center.

Contact Information for Students Support and Learning Resources:
Tel: (00974) 4403 3870
Fax: (00974) 4403 3871
Location: Female Student Activities Building
E-mail: learningcenter@qu.edu.qa

College of Engineering Learning Support

The Engineering Success Oasis (ESO) provides academic support services to all students registered in Engineering courses at QU. ESO provides academic assistance through group and one-on-one tutoring, tailored major programs, and various workshops. Support schedules are announced at the beginning of semesters.

Contact Information for College of Engineering Learning Support:
Females
Tel: (00974) 4403 6380
Email: CENG.SuccessOasis.F@qu.edu.qa

Males
Tel: (00974) 4403 6380
Email: CENG.SuccessOasis.M@qu.edu.qa

Sessions' Booking

Females

1- One-to-one sessions' registration via Simplybook:
https://crulearningfemales.simplybook.me
2- Weekly sessions via email invitation from Engineering Success Oasis (ESO)

Males

1- One-to-one sessions' registration via Simplybook:
https://crulearningmales.simplybook.me
2- Weekly sessions via email invitation from Engineering Success Oasis (ESO)

Student Complaints Policy

Students at Qatar University have the right to pursue complaints related to faculty, staff, and other students. The nature of the complaints may be either academic or non-academic. For more information about the policy and processes related to this policy, you may refer to the student handbook.

Declaration

This syllabus and contents are subject to changes in the event of extenuating circumstances. The instructor (with approval of the Head of Department) reserves the right to make changes as necessary. If changes are necessitated during the term of the course, the students will be notified by email communication and posting the notification on the online teaching tool Blackboard. It is the student's responsibility to check on announcements made while they were absent.

Faculty Name: Dr. Khalil Yousef
Last Modified: 20/08/2026
Date: 20/08/2026